

ABASHIRI, Japan

WMO: 474090

Lat: 44.0178N

Lon: 144.2797E

Elev: 44

StdP: 100.80

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-13.3	-11.7	-18.2	0.8	-11.1	-16.7	0.9	-10.1	12.5	-3.6	11.3	-3.9	2.8	240	0.541

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.9	27.9	21.9	26.0	20.8	24.3	19.8	22.9	26.7	21.8	24.9	20.8	23.5	4.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.6	16.3	25.6	20.6	15.4	24.2	19.7	14.5	23.0	68.4	26.9	64.2	25.0	60.6	23.6	26.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.0	8.4	7.3	-16.1	32.7	1.8	2.3	-17.4	34.4	-18.5	35.7	-19.5	37.0	-20.8	38.6		
(5)			-16.6	24.5	1.7	1.0	-17.8	25.3	-18.8	25.9	-19.8	26.4	-21.1	27.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.0	-5.6	-5.8	-1.4	4.7	10.1	13.7	17.9	19.9	17.1	11.0	4.0	-2.6		
	DBStd	9.64	3.04	3.58	3.63	4.17	4.80	4.03	3.74	3.30	3.00	3.20	3.89	3.21		
	HDD10.0	2115	483	442	353	168	59	10	0	0	0	27	183	389		
	HDD18.3	4281	742	675	611	409	258	146	54	20	59	228	431	648		
	CDD10.0	1021	0	0	0	9	62	122	246	308	212	58	3	0		
	CDD18.3	144	0	0	0	0	3	8	41	70	22	1	0	0		
	CDH23.3	604	0	0	0	6	22	33	176	296	72	1	0	0		
	CDH26.7	128	0	0	0	1	5	4	40	66	12	0	0	0		
(14) Wind	WSAvg	3.2	3.9	3.5	3.6	3.5	3.2	2.6	2.4	2.6	3.0	3.5	3.6	3.8		
(15) Precipitation	PrecAvg	843	61	38	50	51	64	73	84	109	107	82	63	59		
	PrecMax	1209	148	111	141	160	171	209	210	425	299	212	132	176		
	PrecMin	580	11	8	13	8	17	11	11	14	10	22	11	15		
	PrecStd	143	26	22	30	27	33	41	43	69	55	43	28	26		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.5	6.5	11.1	22.1	26.0	26.7	31.0	31.2	28.3	21.5	16.6	9.0		
		MCWB	2.1	3.0	5.8	11.8	15.0	18.4	23.7	24.0	22.1	15.1	11.3	6.3		
	2%	DB	1.6	3.2	7.7	16.6	22.4	23.7	27.1	28.4	25.3	19.2	13.9	5.8		
		MCWB	-0.4	0.4	3.9	9.3	13.5	17.1	21.7	22.9	20.2	14.1	9.8	3.3		
	5%	DB	0.1	1.2	5.8	13.8	19.7	21.7	25.0	26.4	23.2	17.4	11.9	3.8		
		MCWB	-1.3	-1.0	2.3	7.9	13.0	16.3	20.6	21.8	19.0	13.3	8.3	1.7		
	10%	DB	-0.9	-0.5	3.9	11.1	17.3	19.8	23.3	24.7	21.6	15.9	9.8	2.3		
		MCWB	-2.2	-2.2	1.1	6.5	11.9	15.4	19.7	20.9	17.9	12.4	6.9	0.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.3	3.4	6.2	12.7	16.3	19.5	24.1	24.8	22.9	16.9	12.6	6.8		
		MCDB	4.2	5.9	9.7	19.8	23.4	24.5	30.2	30.3	27.0	19.6	15.6	8.6		
	2%	WB	0.1	0.5	4.3	10.2	14.7	18.0	22.4	23.4	21.3	15.2	10.3	3.6		
		MCDB	1.1	2.7	7.5	15.6	20.5	22.6	26.1	27.4	24.1	17.9	12.9	5.3		
	5%	WB	-1.0	-0.8	2.6	8.2	13.5	16.8	21.2	22.3	19.8	14.0	8.6	2.1		
		MCDB	-0.1	1.0	5.4	12.9	18.8	20.6	24.3	25.4	22.0	16.7	11.4	3.5		
	10%	WB	-2.1	-2.0	1.2	6.8	12.2	15.8	20.1	21.4	18.5	12.8	7.1	0.5		
		MCDB	-0.9	-0.5	3.7	10.7	16.8	19.3	22.9	24.2	21.0	15.5	9.7	1.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.0	6.7	6.4	7.6	7.7	6.7	6.0	5.9	6.4	7.2	6.4	5.9		
		MCDBR	7.1	9.2	9.5	13.1	12.5	10.6	9.3	8.6	8.5	8.9	9.8	7.6		
	5% WB	MCWBR	6.1	7.2	6.7	7.3	6.4	5.6	4.9	4.5	4.9	5.7	7.1	6.7		
		MCWBR	6.3	8.5	9.2	12.3	11.2	9.1	8.4	7.9	7.4	7.9	8.9	7.2		
(40) Clear-Sky Solar Irradiance	taub		0.332	0.373	0.445	0.507	0.522	0.499	0.481	0.460	0.408	0.398	0.373	0.344		
		taud	2.074	1.898	1.818	1.828	1.887	2.048	2.166	2.242	2.338	2.218	2.192	2.118		
	Ebn at Noon		750	777	769	760	764	784	794	797	814	757	704	693		
		Edh at Noon	104	149	185	199	193	165	145	130	110	108	93	91		
(44) All-Sky Solar Radiation	RadAvg		1.63	2.65	3.77	4.57	4.88	4.97	4.65	4.23	3.63	2.71	1.69	1.33		
	RadStd		0.14	0.23	0.28	0.32	0.42	0.43	0.51	0.31	0.28	0.13	0.14	0.10		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)			
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page